

Answers

1. **B** 2. **11** 3. **C** 4. **6** 5. **84** 6. **60** 7. **D** 8. **40**
9. **D** 10. **D** 11. **A** 12. **C**

Explanations

1. **B**

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now taking average processing time per order for the above mentioned categories we get :

$$\text{Art} = \frac{2+8+2+1+7}{5} = 4$$

$$\text{Binders} = \frac{1+1+11+2}{4} = 3.75$$

$$\text{Papers} = \frac{3+2+5}{3} = 3.33$$

$$\text{Phones} = \frac{2+12+1}{3} = 5$$

So in decreasing order we get Phones , Art , Binder , Paper.

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2. **11**

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 ,5-16
Paper	x	4-7	2-4 ,7-12
Phones	2-4 ,5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now Envelopes and Accessories has only 1 ship mode i.e Standard class .

So therefore processing days for envelopes = 7-3 =4

and processing days for accessories = 19-1 =18

Therefore average = $\frac{(18+4)}{2} = 11$

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3.C

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 ,5-16
Paper	x	4-7	2-4 ,7-12
Phones	2-4 ,5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now from the table we observe that the total number of orders are 35 and 7 orders have a processing time of 1 unit

The 7 orders are : Arts Standard class, Binders First class and standard class, Phones First class, Bookcases second class (2 orders) and Chairs standard class.

So the percentage = $\frac{7}{35} \times 100 = 20$

4.6

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now No booking days from the table are : September 8,9,10,11,12 and 14.

So a total of 6 days .

5.84

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right) a = \frac{5}{4} a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8} a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right) b = \frac{3}{4} b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16} b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right) c = \frac{4}{5} c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25} c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right) d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right) e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right) f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of b-f = 60, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: b = 64



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6.60

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	b	1	$3b/4$		$9b/16$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	e = 60	3	$5e/4 = 75$		$3e/2 = 90$	
F	f	6	$2f$	6	$3f/2$	6

SI of E in 2016 is 60.

Hence, the correct answer is 60.

[VIEW SOLUTION](#)

7. D

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right) e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right) f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of b-f = 60, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: b = 64

If b = 64, then the value of f becomes 4, as we know b - f = 60.

If $f = 4$, then the value of F in 2024 is $3f/2 = 12/2 = 6$.

We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: $b = 96$

If $b = 96$, then the value of f becomes 36, as we know $b - f = 60$.

If $f = 36$, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 * 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B .

So, we can eliminate this case.

CASE 3: $b = 80$

If $b = 80$, then the value of f becomes 20, as we know $b - f = 60$.

If $f = 20$, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 * 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of $b = 80$ and $f = 20$.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	$b = 80$	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	1
F	$f = 20$	6	$2f = 40$	6	$3f/2 = 30$	6

So, SI of B in 2024 is 45.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

8.40

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A :

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	$b = 80$	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	1
F	$f = 20$	6	$2f = 40$	6	$3f/2 = 30$	6

So, SI of F in 2020 is 40.

Hence, the correct answer is 40.

[VIEW SOLUTION](#)

9. D

The total project month is the number of months Abani, Bahni, Danni, and Tinni individually worked for all the projects combined :

Abani - $2+2+5 = 9$ months

Bahni - $2+4+3 = 9$ months

Danni - $3+3+2+1 = 9$ months

Tinni - $2+2+3+2 = 9$ months.

The total employee month for all the five projects is the sum of the total employee-month is the sum of the number of months four employees worked in this project.

Project -1 = $2+2+2 = 6$ months

Project -2 = $3+2 = 5$ months

Project - 3 = $2+4+3 = 9$ months.

Project - 4 = $5+2+3 = 10$ months.

Project - 5 = $3+1+2 = 6$ months.

Only statement 1 is true.

[VIDEO SOLUTION](#)

10. D

Considering the information provided :

For project 1 : 3 months.

Project - 2: 3 months.

Project - 3: 5 months.

Project - 4: 5 months.

Project - 5: 4 months.

Among the given options option D is true which is project 3, project 4.

 VIDEO SOLUTION



11. A

Abani, Banni, and Danni did not work on multiple projects simultaneously in a month

Tinni was the only person who worked on multiple projects which are project 4 and project 5 in the month of september.

 VIDEO SOLUTION

12. C

The annual completion index for different people is :

The weightage average of the completion percentage assigned from the different projects, with the weights being the corresponding number of months (s)he worked in these projects.

For Abani :

$$\frac{((100 \cdot 2) + (100 \cdot 2) + (80 \cdot 5))}{2 + 2 + 5} = \frac{800}{9}$$

For Bahni :

$$\frac{((100 \cdot 2) + (75 \cdot 4) + (90 \cdot 3))}{2 + 3 + 4} = \frac{770}{9}$$

For Danni :

$$\frac{((90 \cdot 3) + (100 \cdot 3) + (100 \cdot 2) + (100 \cdot 1))}{2 + 3 + 2 + 1} = \frac{870}{9}$$

For Tinni :

$$\frac{((80 \cdot 2) + (100 \cdot 2) + (100 \cdot 3) + (100 \cdot 2))}{2 + 2 + 3 + 2} = \frac{860}{9}$$

The descending order for the four people is :

Danni, Tinni, Abani, Bahni.

 VIDEO SOLUTION